

nology and is compatible with the industry-standard MCS-51 instruction set. By combining a versatile 8-bit CPU with Flash on a monolithic chip, the Atmel AT89C2051 is a powerful microcomputer that provides a highly flexible and cost-effective solution to many embedded control applications.

The AT89C2051 features 2 kB of Flash memory, 128 bytes of RAM, 15 input/output (I/O) lines, two 16-bit timers/counters, five-vector two-level interrupts, full-duplex serial port, precision analogue comparator, on-chip oscillator and clock circuitry. It is designed with static logic for operation down to zero frequency and supports two software-selectable power-saving modes. The idle mode stops the CPU while allowing the RAM, timers/counters, serial port and interrupt system to continue functioning. The power-down mode saves the RAM contents but freezes the oscillator, disabling all other chip functions until the next hardware reset.

Circuit description

Fig. 1 shows the circuit of the *pranayam* timer based on microcontroller AT89C2051. Port pins P1.0 through P1.7 of the microcontroller are connected to D0 through D7 for driving data lines of the LCD module. We are not reading the status of the LCD, i.e., whether the LCD is busy or not, hence R/W pin is directly grounded. Instead of reading the status of the LCD, we use a fixed delay before sending command or data to the LCD. The register-select (RS) pin is connected to port pin P3.0 and Enable pin is connected to port pin P3.1. Preset VR1 is used to control the contrast of the LCD.

A 12MHz crystal connected at pins 4 and 5 of the microcontroller generates the basic clock frequency. Power-on reset is derived from the combination of capacitor C6 and resistor R2. Switch S2 is used for manual reset.

The user (*sadhak*) can set the

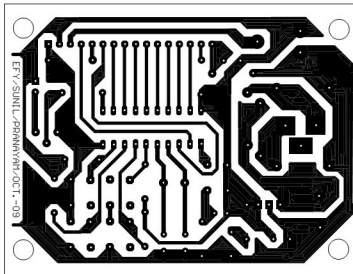


Fig. 2: A single-side, actual-size PCB layout for microcontroller-based pranayam timer

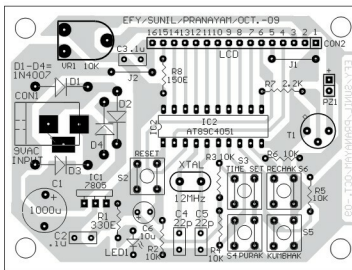


Fig. 3: Component layout for the PCB

time for *pranayam* using switches S3 through S6. Switch S3 is connected to external interrupt pin INT0 (port pins P3.2) for initiating the process of time setting for *purak*, *kumbhak* and *rechak*. Switches S4, S5 and S6 are used to set the time durations for *purak*, *kumbhak* and *rechak*. Port pin P3.7 drives piezobuzzer PZ1 through transistor T1.

After power-'on' reset (or manual reset), the LCD shows the message "Pranayam Timer" in the first line and "Purak: 04" in the second line. This means the count value for the *purak* process is '04,' which decrements by one every second. When it reaches '01,' the buzzer beeps and the *kumbhak* countdown starts with message

"Kumbhak: 16." The *kumbhak* process should be four times longer than *purak*, so the countdown automatically starts from the value '16.' After completion of *kumbhak*, the buzzer beeps again and the *rechak* countdown starts with message "Rechak: 08." As *rechak* is two times longer than *purak*, the countdown starts from '08.' The cycle of *purak*, *kumbhak* and *rechak* countdown repeats continuously until you turn off the *pranayam* timer device.

When you press switch S3 once after pressing reset switch S2, external interrupt 0 generates and messages for time setting for *purak*, *kumbhak* and *rechak* appear on the LCD one by one. Switch S3 starts the process of time set-

ting for *purak*, *kumbhak* and *rechak*.

Press switch S4 for setting the time for *purak* after message "Set Purak Time" appears in the first line of the LCD. On releasing switch S4, message "Set Kumbhak Time" appears on the LCD. Set the time for *kumbhak* by pressing switch S5. On releasing switch S5, message "Set Rechak Time" appears on the LCD. Set the time for *rechak* by pressing switch S6. On releasing switch S6, message "Pranayam Timer" appears in the first line of the LCD and the countdown for *purak*, *kumbhak* and *rechak* starts with new settings.

In order to derive the power supply for the circuit, 230V AC mains is stepped down by transformer X1 to